THE CATEGORY OF ARROWS FROM A SUBCATEGORY IS FILTERED

THE STACKS PROJECT AUTHORS

Allowed standard prerequisites (editorial).

- Filtered categories. categories-definition-directed (cross-chapter, named). Defines the conditions checked.

Source reference. Original reference 002V: focused-seeds/geometry/S9-eqv-maps-from-compact-filtered/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a triangulated category. Let $\mathcal{D}' \subset \mathcal{D}$ be a full triangulated subcategory. Let $X \in \mathrm{Ob}(\mathcal{D})$. The category of arrows $E \to X$ with $E \in \mathrm{Ob}(\mathcal{D}')$ is filtered.*

Proof. We check the conditions of Categories, Definition 002V. The category is nonempty because it contains $0 \to X$. If $E_i \to X$, $i = 1, 2$ are objects, then $E_1 \oplus E_2 \to X$ is an object and there are morphisms $(E_i \to X) \to (E_1 \oplus E_2 \to X)$. Finally, suppose that $a, b : (E \to X) \to (E' \to X)$ are morphisms. Choose a distinguished triangle $E \xrightarrow{a-b} E' \to E''$ in $\mathcal{D}'$. By Axiom TR3 we obtain a morphism of triangles

$$\begin{array}{ccccc} E & \xrightarrow[a-b]{} & E' & \longrightarrow & E'' \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & X & \longrightarrow & X \end{array}$$

and we find that the resulting arrow $(E' \to X) \to (E'' \to X)$ equalizes a and b. $\square$

Date: stacks-project:equiv.tex:local-snapshot-20261012; GNU FDL 1.2 or later.